UNIVERSITY OF OXFORD

DEPARTMENT OF PAEDIATRICS

METHODOLOGY, EVIDENCE SYNTHESIS, AND IMPLEMENTATION | MESISIG
IASP Special Interest Group

Navigating AI in Evidence Synthesis

AI tools across the review workflow

Dr Luke Baxter

Principal Scientist · Department of Paediatrics · University of Oxford

Map and presentation

lukebax.github.io/ai-review-tools/

Stages of a systematic review

1 Protocol

2 Search

3 Screening for relevance

4 Quality assessment

5 Data extraction

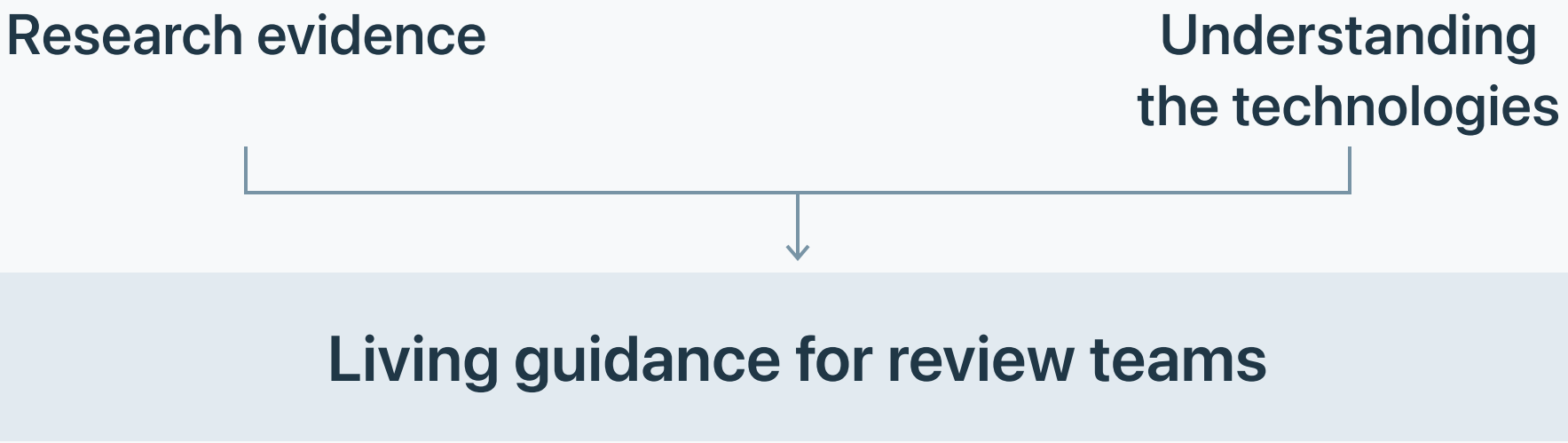
6 Data synthesis

7 Certainty of evidence

8 Manuscript

RAISE provides guidance for using AI in reviews

Responsible use of AI in evidence synthesis

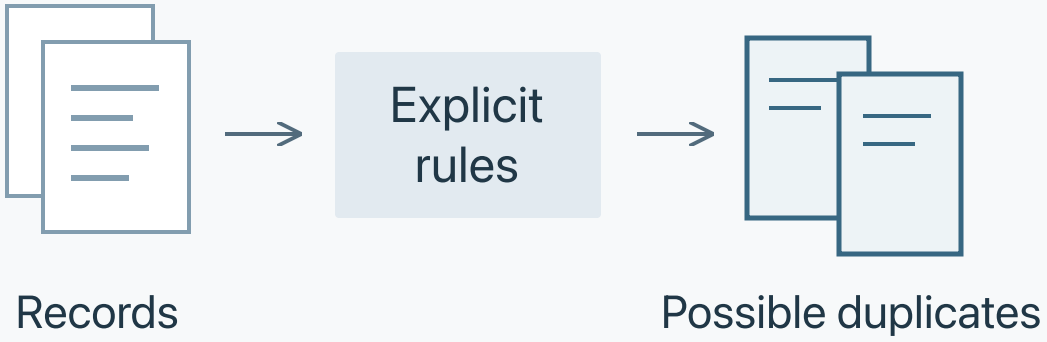


Developed by researchers working in evidence synthesis

Explicit rules and patterns in unlabelled data

Rule-based algorithms

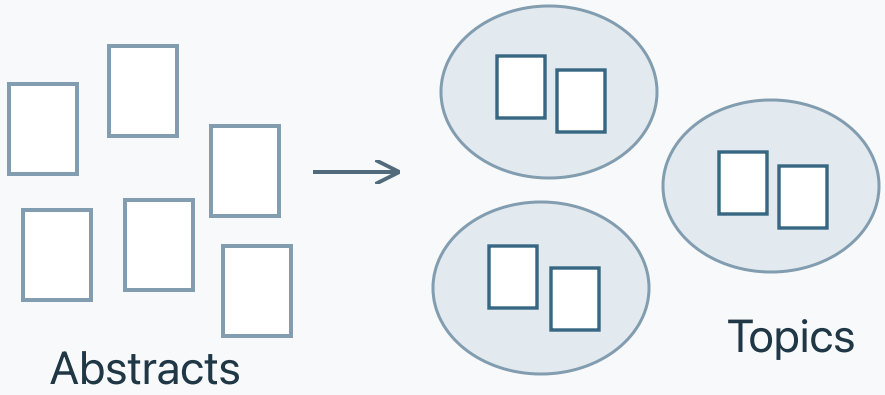
Deduplication



Predefined rules compare the records.

Unsupervised machine learning

Topic modelling

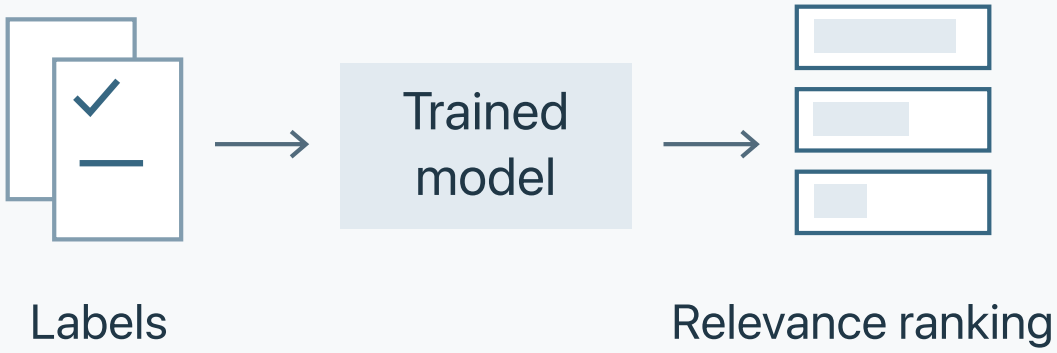


Patterns emerge without supplied labels.

Learning from labelled examples and generating language

Supervised machine learning

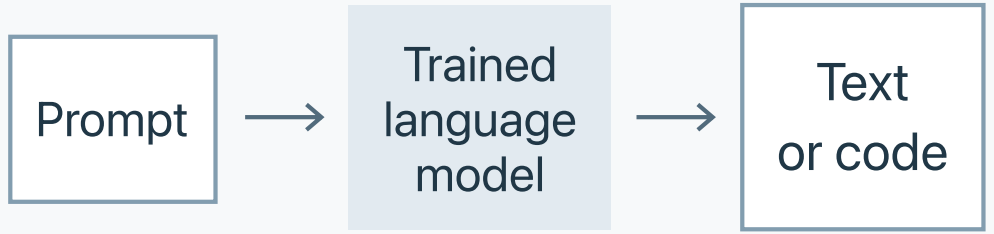
Prioritised screening



Already trained or trained during the review

Generative LLMs

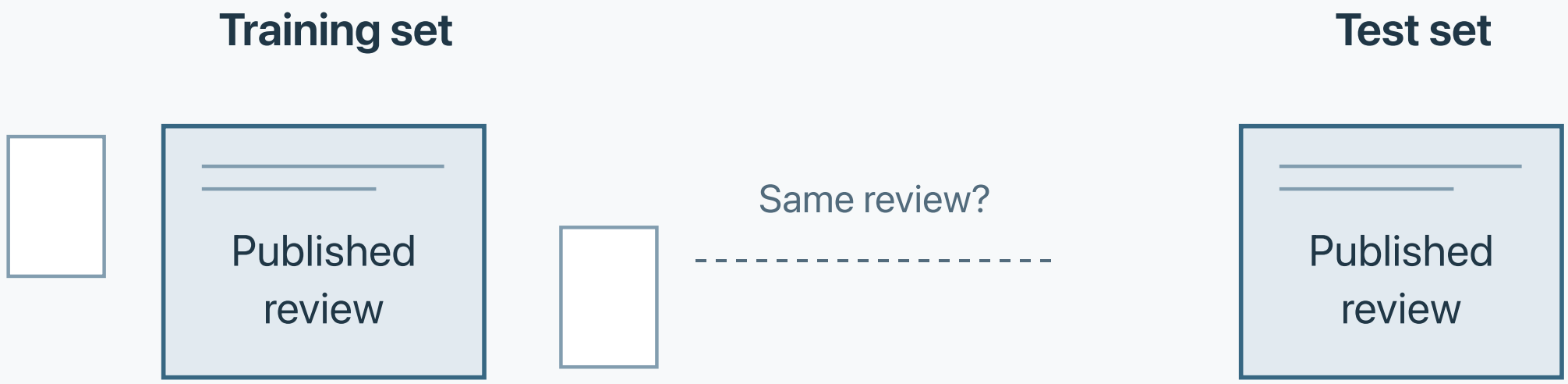
Writing analysis code



Outputs can vary.
Models can hallucinate.

Evaluation needs independence from development

Possible training–test overlap: a hypothetical example



Apparent performance may overstate performance on new material.

Incomplete disclosure makes overlap harder to establish.

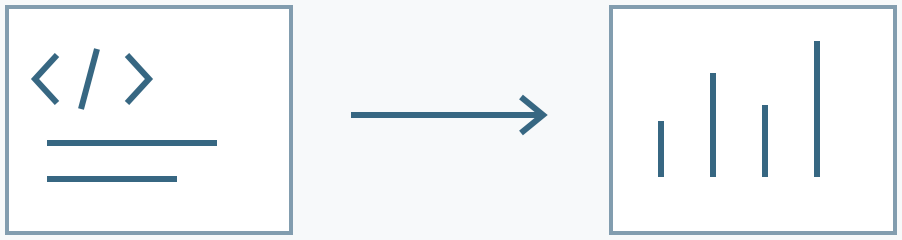
Recommendations apply to a particular use

Acceptable for use	Use outputs, accounting for limitations and bias
Human verification required	Check outputs and correct errors
Requires validation within the review	Test performance against a prespecified threshold and suitable benchmark
Exploratory and supplementary use	Explore or supplement established methods
Not acceptable for use	Do not rely on outputs for this task

Verification and validation ask different questions

Verification

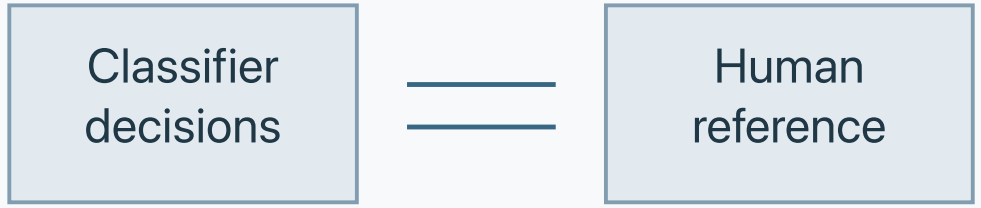
Check the output



Does it carry out the intended analysis?

Validation

Formally evaluate performance

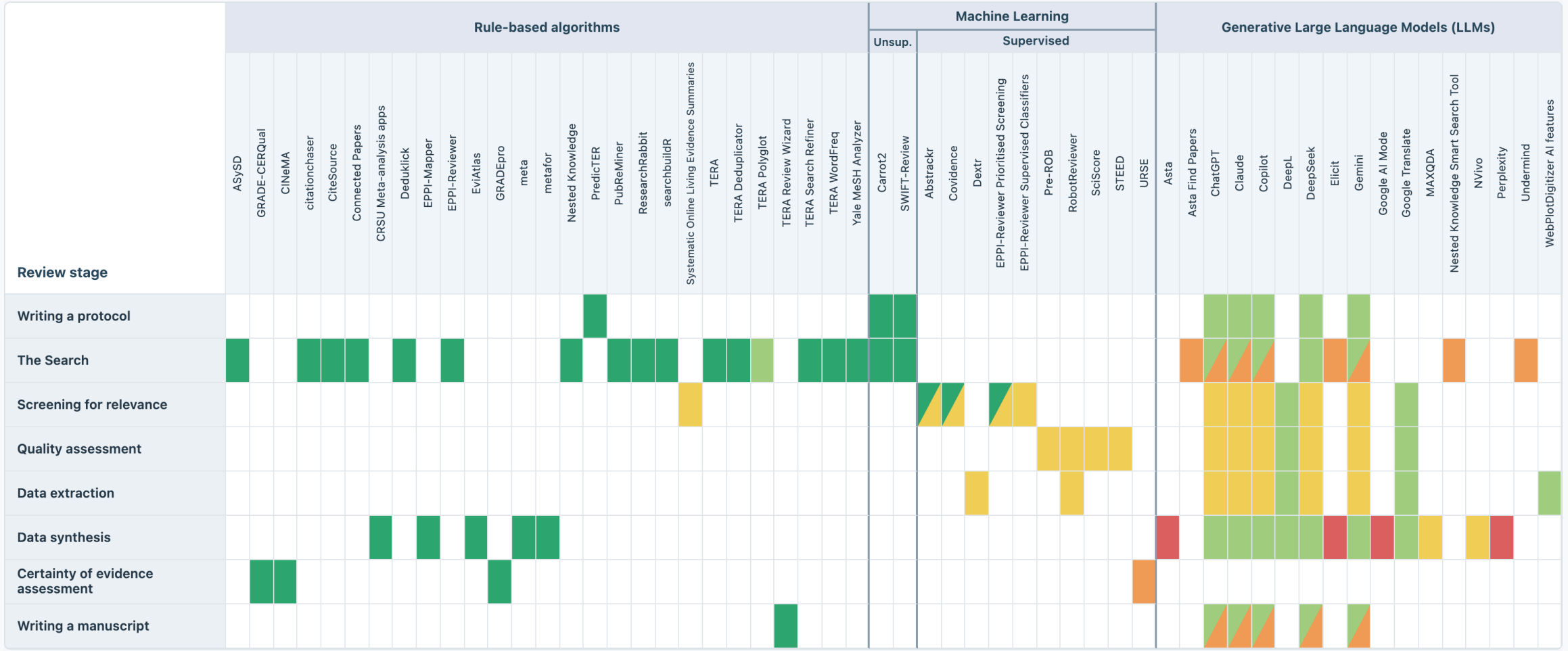


Do relevance judgements meet the prespecified standard?

Use robust, relevant evidence. Follow any requirement for validation within the review.

A workflow may need both.

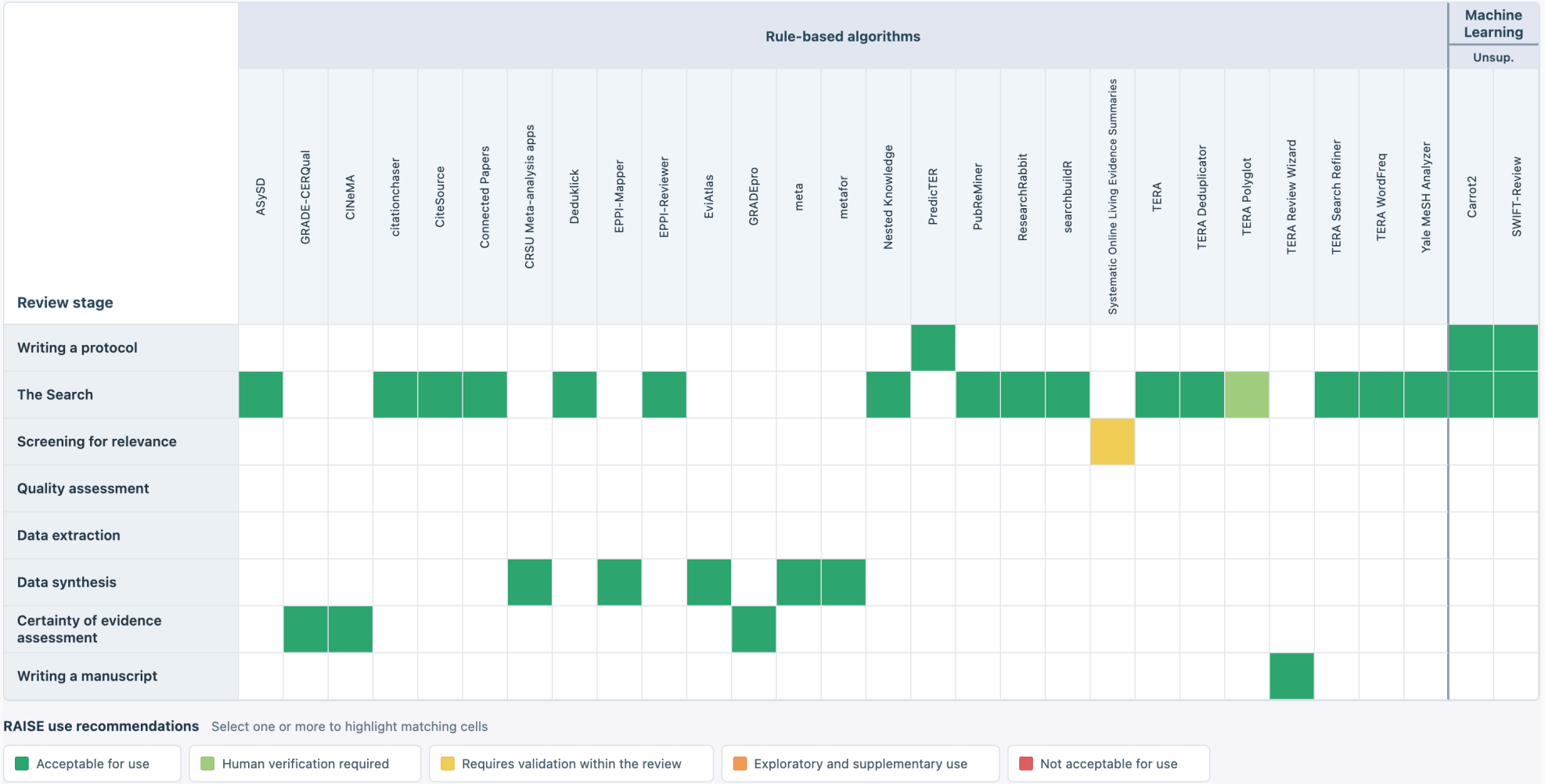
AI tools across the review stages



Find interactive map here: lukebax.github.io/ai-review-tools/

Rule-based algorithms and unsupervised machine learning

Most of these mapped uses are acceptable



Find interactive map here: lukebax.github.io/ai-review-tools/

Supervised machine learning

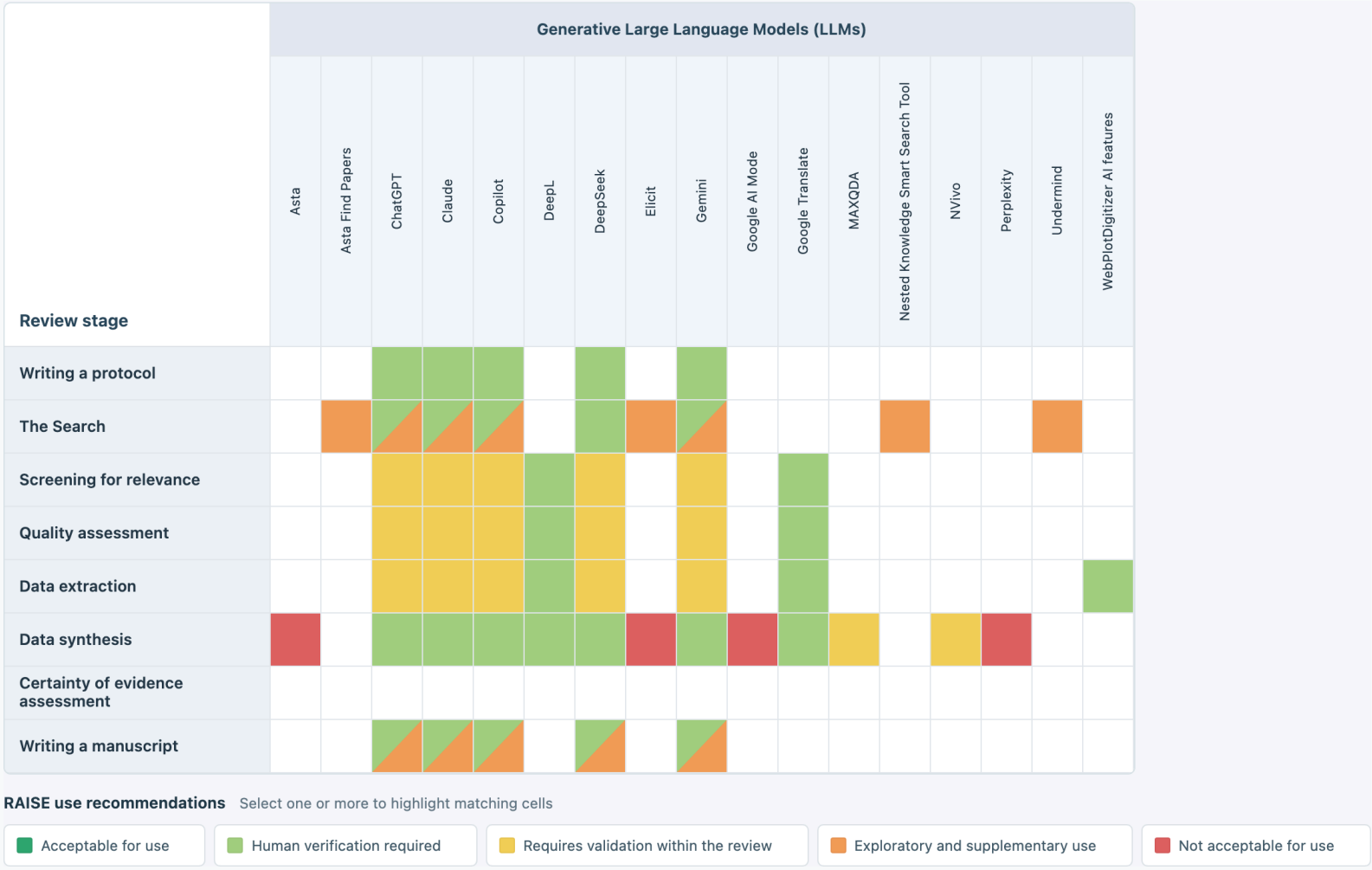
Most mapped uses need validation: performance depends on task and material



Find interactive map here: lukebax.github.io/ai-review-tools/

Generative large language models (LLMs)

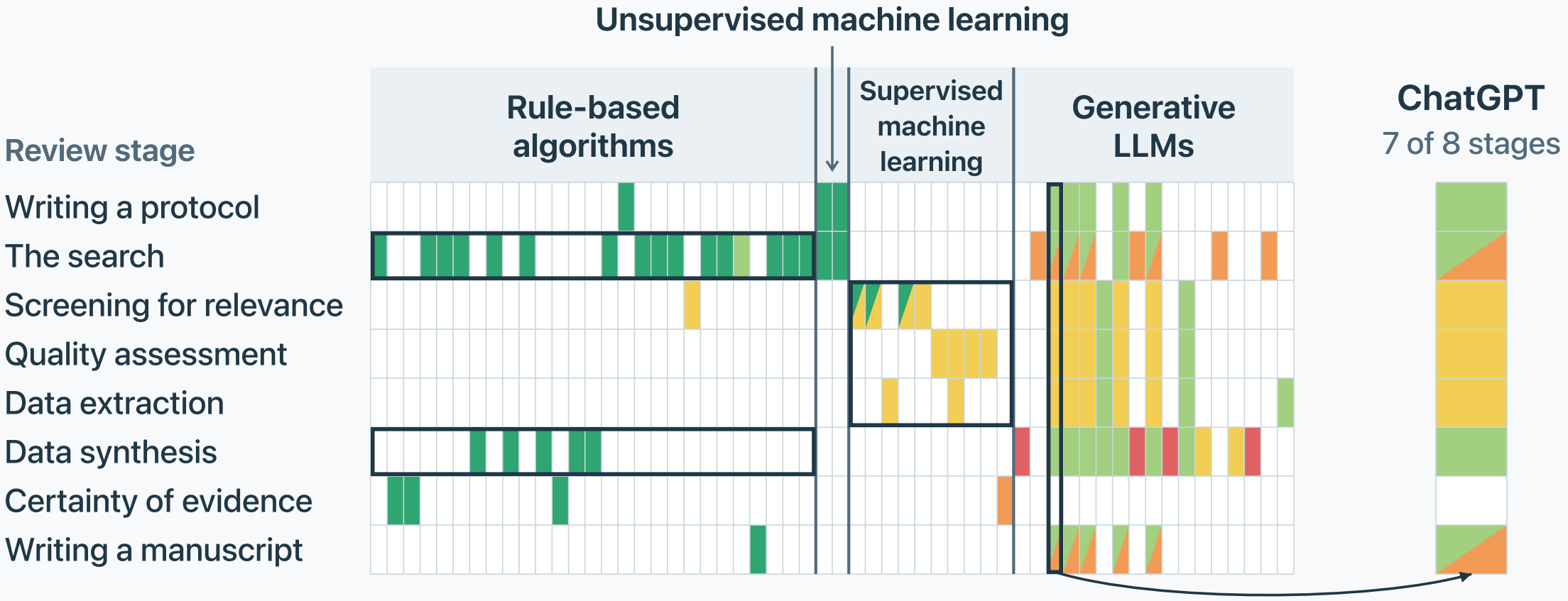
LLMs can hallucinate



Find interactive map here: lukebax.github.io/ai-review-tools/

Tools differ in where and how broadly they are used

Stage clusters in the overview, with ChatGPT enlarged alongside



Rule-based
Record matching and calculations

Supervised
Relevance, methods and data

ChatGPT
Language and code across stages

Find interactive map here: lukebax.github.io/ai-review-tools/

Prioritised screening with and without early stopping

Stopping early changes reliance on the ranking

Abstrackr



Covidence



EPPI-Reviewer Prioritised Screening



Screen every record

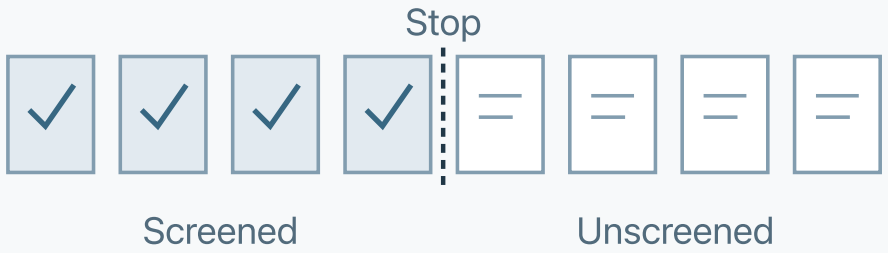


Every record is screened

Acceptable for use

A poor ranking does not leave records unseen.

Stop early



Requires validation within the review

Assess the screening and stopping procedure.

The ranking can be identical. What changes is how much we rely on it.

Two different uses during screening

Screening for relevance

Google Translate

Translation support

Human verification required

Check the translation against the original.

Google’s Gemini

Relevance judgements

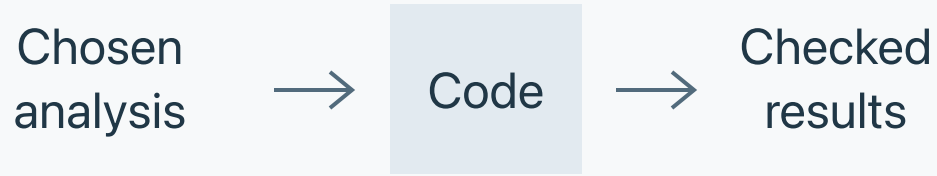
Requires validation within the review

Evaluate the screening procedure’s performance.

Selected screening cells from the map.

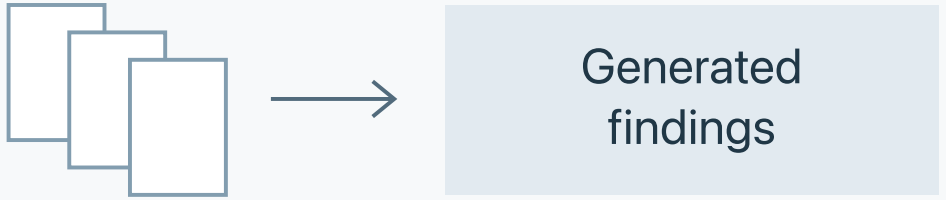
Synthesising findings differs from drafting analysis code

LLM-drafted code for a chosen analysis



Human verification required

LLM-derived synthesis across studies



Not acceptable for use in a formal review

Qualitative coding or themes: validation and verification.
The boundary with delegated synthesis is not fully defined.

RAISE also restricts complete agentic reviews

Complete agentic review: not acceptable for use



Using an LLM agent for one review task requires its own assessment.

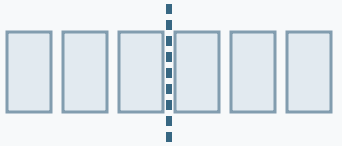
Tool suitability

Prioritised screening with early stopping

1

What is the purpose of the tool?

Assess ranking plus the stopping procedure.



2

Where have the training, testing and validation data come from?

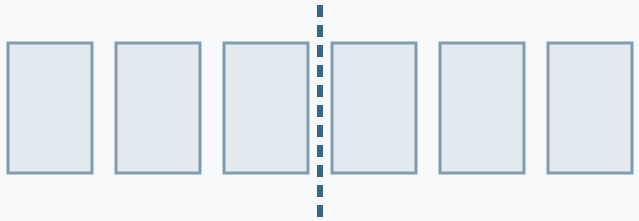
Training labels: quality and selection.
Testing and validation data: independence and relevance.



Validation and performance

Prioritised screening with early stopping

3 Is the tool validated and performing sufficiently well for use?



Compare performance

Human-screened reference

Performance of the stopping procedure

Set the performance standard in advance.
Establish whether it is met within this review.

An appropriate human-screened reference

Team, resources and documentation

Prioritised screening with early stopping

4 **Does the team have the expertise and resource to use the AI tool effectively, and is the usability acceptable?**

Skills, assessment time and a workable process

5 **Is there sufficient transparency around licenses and other available documentation?**

Instructions, access, data handling, version and settings

Missing stopping evidence must be resolved before reliance.

Choosing a suitable use of AI

Consider how the tool works, the task and the evidence needed before relying on its outputs.



RAISE provides the guidance.

The companion map helps you explore RAISE's recommendations and the tools.

Map and presentation

lukebax.github.io/ai-review-tools/